

Detect, Recover, Interpret: Black-Box Auditing of Hidden Routing Regimes

Revised algorithm note (v2): calibrated detection and certified recovery

Working algorithm note

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Abstract

This revision restates the detection target and hardens the machinery of the three-stage auditing framework. The principal changes relative to v1 are: (1) the Stage A null is reframed — the tested hypothesis is compatibility with a *calibrated smooth single-branch reference distribution*, not the literal statement $K = 1$; rejection is evidence of boundary-like multiscale response geometry, consistent with latent routing subject to corroboration; (2) the ratio statistic R_i is replaced by a stabilized log-range statistic and, preferably, a studentized deviation from a fitted smooth-null scale curve, with a minimum-signal abstention rule; (3) the null generator $\hat{\mathcal{P}}_0$ is made concrete as a hierarchical Gaussian-process model of single-branch local response surfaces, fitted robustly and cross-fitted for testing; (4) Benjamini–Hochberg screening is demoted to a screening device; the primary confirmatory claim is an omnibus selection-aware parametric bootstrap whose p -value has standard finite-simulation form and whose calibration is conditional on generator adequacy; (5) shape corroboration specifies its scalar projections explicitly; (6) Stage B uses whitened features and soft mixture responsibilities, commits to a concrete partial-matching label-synchronization algorithm, fits gates by regularized responsibility-weighted logistic regression with stage-repeating bootstrap uncertainty, and replaces propagation through unflagged regions with bridge-probe *edge certification*, on the principle that non-rejection is not evidence of interiority; (7) the audit design adds budget splitting, negative controls, and probe-support diagnostics; (8) Stage C interpretation language is weakened to what the evidence supports. A closing section states what Stage B still requires to reach Stage A’s level of maturity.

1 Problem, target, and the precise detection claim

Let a black-box system be represented as

$$f(x) = f_{g(x)}(x) + \eta,$$

where $g : \mathcal{X} \rightarrow [K]$ is an unobserved routing rule, f_k is the response mechanism in regime k , and η is observation or sampling noise. The auditor observes only queries and responses. For anchors $x_1, \dots, x_n$, a probe family $Q_\sigma(\cdot \mid x_i)$ generates perturbations

$$\delta_{ij}^{(t)} \sim Q_{\sigma_t}(\cdot \mid x_i), \quad \psi_{ij}^{(t)} = \psi\left(f(x_i + \delta_{ij}^{(t)})\right) \in \mathbb{R}^V,$$

with response feature map $\psi : \mathcal{Y} \rightarrow \mathbb{R}^V$. The local routing composition is $\pi_i^\sigma(k) = \mathbb{P}_{\delta \sim Q_\sigma}[g(x_i + \delta) = k] \in \Delta^{K-1}$. The three goals remain separated: **detection** (Stage A), **recovery** (Stage B), and **interpretation** (Stage C).

1.1 What Stage A does and does not test

A single smooth mechanism can produce curvature-driven scale dependence, spatially varying derivatives, heteroskedastic output noise, anisotropic noise, sharp-but-continuous transitions, and unstable normalization at locally flat anchors. Conversely, genuine routing can remain invisible when branches overlap under ψ , when probes move parallel to the gate, or when the local mixture is effectively unimodal. Stage A therefore does not, and cannot, test the literal hypothesis $K = 1$. The tested null is

$$H_{0,i} : \text{anchor } i \text{ follows the calibrated smooth single-branch null } \hat{\mathcal{P}}_0,$$

and the operational claim is:

Given a specified response representation ψ , probe design $\{Q_{\sigma_i}\}$, and null class, identify anchors whose multiscale response geometry is incompatible with the fitted reference distribution of single-branch smooth behavior.

Rejection is *consistent with* boundary crossing or latent routing and must be corroborated by shape evidence (Stage A.5), recovery structure (Stage B), and mechanism analysis (Stage C) before any routing language enters the audit finding. Both positive and negative findings are relative to ψ , the probe design, and the null class: a sharp-but-continuous transition narrower than the probed scales is observationally equivalent to a routing boundary, and a mechanism switch invisible through ψ is undetectable in principle.

2 Audit design and budget discipline

2.1 Budget split

Partition the query budget into four disjoint samples:

- $\mathcal{S}_1$ — detection and scale selection (Stage A.1–A.4);
- $\mathcal{S}_2$ — modality confirmation at flagged anchors (Stage A.5);
- $\mathcal{S}_3$ — mixture recovery, synchronization, and gate fitting (Stage B);
- $\mathcal{S}_4$ — final held-out validation of every reported quantity.

The selection-aware bootstrap can in principle account for sample reuse, but a genuine held-out split is easier to defend and removes the dependence of confirmatory claims on correctly modeling the reuse. When the budget forces reuse, the bootstrap must reproduce it exactly (§3.6).

2.2 Probe families and support diagnostics

Maintain at least two probe families: *on-manifold* (perturbations consistent with the data distribution) and *ambient* (e.g., isotropic Gaussian in input space), across a ladder $\sigma_1 < \dots < \sigma_T$; feature-targeted probes may be added when a candidate gating feature exists. For every anchor and probe family, report a *support realism score* for the perturbed inputs — e.g., the quantile of their k NN distance to a reference sample of real inputs. Findings are reported stratified by support realism: “ambient-only” evidence obtained at unrealistic inputs is explicitly qualified, since it may reflect behavior in operationally meaningless regions.

2.3 Negative controls

Every audit run includes controls probed with the identical pipeline: smooth curved single-regime functions; heteroskedastic single-regime functions; discontinuous functions without latent routing (e.g., a single mechanism containing a hard kink); known benign mixtures of experts; and randomized or irrelevant probe directions. Controls establish whether the statistic responds specifically to boundary-like switching rather than to generic nonlinearity or noise structure. The discontinuous control also makes the observational-equivalence point of §1.1 concrete: an intrinsic step inside one mechanism *will* be flagged, because at the level of observable behavior it is a regime boundary; attributing mechanism is Stage C’s job, not Stage A’s.

3 Stage A: multi-scale detection

3.1 A.1: Noise model, whitening, and the dispersion curve

Repeated unperturbed queries at each anchor (and pooled across anchors) yield an estimated observation-noise covariance $\hat{\Sigma}_{\text{obs}} \in \mathbb{R}^{V \times V}$; when V is large relative to the number of repeats, use a shrinkage or low-rank estimator (e.g., Ledoit–Wolf). Whiten all response features:

$$\tilde{\psi}_{ij}^{(t)} = \hat{\Sigma}_{\text{obs}}^{-1/2} \psi_{ij}^{(t)}.$$

Why this step: scalar noise-floor subtraction implicitly assumed isotropic noise; whitening makes the floor isotropic by construction, at the cost of requiring a stable covariance estimate.

With $\tilde{S}_i(\sigma_t)$ the empirical scatter of $\{\tilde{\psi}_{ij}^{(t)}\}_{j=1}^m$, the repeated-query scatter is $\approx I$, so the excess top-eigenvalue spread per unit input scale is

$$r_i(\sigma_t) = \frac{\sqrt{(\lambda_{\max}(\tilde{S}_i(\sigma_t)) - 1)_+}}{\sigma_t}.$$

For a locally linear single branch with whitened directional slope b_i , $\lambda_{\max}(\tilde{S}_i(\sigma)) \approx 1 + \sigma^2 \|b_i\|^2$ and $r_i(\sigma) \approx \|b_i\|$; curvature makes r_i rise with σ even under the null, which is why the null model must represent curvature (§3.3) rather than the statistic assuming linearity.

Minimum-signal abstention. If $\lambda_{\max}(\tilde{S}_i(\sigma_t)) - 1$ is not significantly positive at any scale (one-sided test against the repeated-query eigenvalue distribution), the anchor is reported as *insufficient signal*, which is distinct from *null-consistent*: a locally flat model provides no information about boundaries, and a noise-dominated denominator must not enter any ratio or range statistic.

Why this step: the v1 ratio could explode when one scale’s spread was noise-underestimated; the anchor looked anomalous for reasons unrelated to any bump.

3.2 A.2: Stabilized scale statistics

The v1 ratio $R_i = \max_t r_i / \min_t r_i$ is deprecated: its denominator is unstable near flat regions, and its null distribution conflates bump detection with normalization noise. Two replacements, in increasing order of preference:

S1. Log-range.

$$R_i^{\text{log}} = \max_t \log(r_i(\sigma_t) + \epsilon) - \min_t \log(r_i(\sigma_t) + \epsilon),$$

with a fixed small $\epsilon > 0$, computed only on anchors passing the minimum-signal condition. A robust default when no null generator is yet fitted.

S2. Studentized deviation from the fitted null scale curve. With the null generator of §3.3 fitted, obtain by simulation (or GP analytics) the null mean and standard deviation of the log dispersion curve at anchor i , $\mu_{0,i}(\sigma_t)$ and $s_{0,i}(\sigma_t)$, and set

$$T_i^{\text{scale}} = \max_t \frac{\log r_i(\sigma_t) - \mu_{0,i}(\sigma_t)}{s_{0,i}(\sigma_t)}, \quad \sigma_i^* = \arg \max_t \frac{\log r_i(\sigma_t) - \mu_{0,i}(\sigma_t)}{s_{0,i}(\sigma_t)}.$$

This asks the calibrated question directly: *is the observed scale dependence larger than the fitted population of smooth single-branch behaviors predicts at this anchor?*

Why this step: a bump statistic should measure deviation from the null’s own scale dependence, not deviation from flatness; curvature is a null feature, not evidence.

3.3 A.3: The null generator $\hat{\mathcal{P}}_0$

The null generator is the object on which all calibrated inference rests, so it is specified concretely. $\hat{\mathcal{P}}_0$ is a fitted joint generative law over a *complete synthetic audit dataset* — all responses at all anchors, scales, and probes, plus repeats — under the single-branch hypothesis. It must represent: (a) smooth local response surfaces with realistic slopes and curvatures; (b) cross-anchor variation in those slopes and curvatures; (c) observation noise, including heteroskedasticity across anchors; (d) spatial or batch dependence among anchors.

Hierarchical GP specification (default). At anchor i , model each whitened response coordinate (or each of the leading response principal components) of the local surface as a Gaussian process in the perturbation:

$$m_i(\delta) \sim \mathcal{GP}\left(\beta_i^\top \delta, s_i^2 k_\nu(\|\delta - \delta'\|/\ell_i)\right), \quad \tilde{\psi}_{ij} \mid m_i \sim \mathcal{N}\left(m_i(\delta_{ij}), \Sigma_i^{\text{noise}}\right),$$

with Matérn kernel k_ν . The hyperparameters have direct null meaning: β_i is the local linear slope, s_i the amplitude of smooth nonlinearity, ℓ_i the curvature lengthscale, and ν the differentiability class — “how smooth is smooth” is an explicit modeling choice rather than an implicit one. Cross-anchor structure is hierarchical: $(\log s_i, \log \ell_i, \beta_i)$ are drawn from population laws fitted across anchors, with spatially smooth mean fields when anchors are spatial; Σ_i^{noise} comes from local repeats or a fitted variance function. Simulation is direct: draw hyperparameters, draw the surface values at the actual probe locations (a finite-dimensional Gaussian with the kernel Gram matrix), add noise — one synthetic audit per replicate, at the original anchors, ladder, and probe counts, with spatial dependence simulated jointly by blocks.

Fitting under contamination. The null must be fitted from data that may contain alternatives. Options, in increasing order of strength: trimmed fitting that excludes anchors in the top fraction of a preliminary detection statistic (report sensitivity to the trimming fraction); a separate calibration anchor set never used for testing; and **cross-fitting** — partition anchors into folds, test each fold against a generator fitted on the remaining folds. Cross-fitting is the default for anchor-level screening p -values.

Why this step: fitting $\hat{F}_0$ or $\hat{\mathcal{P}}_0$ on the same statistics being tested makes the null random, contaminated, and dependent on every observation; empirical-null BH over such fits has no automatic FDR guarantee.

Misspecification honesty. The generator’s smoothness class *defines* the null of §1.1. A single mechanism with a transition sharper than ℓ_i at the probed scales is outside the null and will be flagged; that is the intended behavior, and the finding is stated as boundary-like geometry, not as proof of a latent gate. Calibration of every downstream p -value is conditional on generator adequacy; the evaluation plan (§12) therefore includes misspecification experiments (heavy-tailed noise, unmodeled heteroskedasticity, kinks, wrong ν) measuring realized error rates when the generator is wrong. Alternative null classes (local polynomial with random coefficients, splines) are acceptable substitutes; the GP default is chosen because smoothness is the null hypothesis and the GP parameterizes exactly smoothness with few, interpretable, estimable hyperparameters.

3.4 A.4: Screening and multiplicity

Compute anchor-level screening p -values p_i for T_i^{scale} (or R_i^{log}) by simulation from the cross-fitted generator. Apply Benjamini–Hochberg at rate q to obtain the flagged set F . In this framework BH is a *screening device* that concentrates the confirmation budget; it is not reported as a final FDR guarantee, because the fitted null is estimated and anchors may be dependent. Dependence is addressed by blockwise simulation in the generator, block-level testing, or Benjamini–Yekutieli as sensitivity analyses. Formal error control lives in the omnibus bootstrap claim (§3.6); anchor-level confirmatory claims additionally require the per-anchor selection-aware p -values with a dependence-robust multiplicity correction.

Why this step: screening and confirmation are different jobs; conflating them forced v1’s empirical-null BH to carry a calibration burden it could not meet.

3.5 A.5: Shape corroboration with specified projections

A dispersion bump is corroborated by evidence of multiple response populations. Hartigan’s dip test is univariate, so the scalar projection must be specified:

- P1. Default:** dip on the first principal-component score of the whitened confirmation sample ($\mathcal{S}_2$) at σ_i^* .
- P2. Prespecified set:** dip on a small fixed set of directions — top two PCs plus, when available, a candidate gating direction — with Bonferroni correction over the set.
- P3. Projection pursuit (optional):** maximize a bimodality index over directions; if used, the entire pursuit is repeated inside every bootstrap replicate.

Known power limitation. Principal components maximize variance, not between-regime separation; strong within-regime anisotropy can hide a mixture from the top PC. A non-significant dip therefore does not certify unimodality. Multivariate modality or clustering-based tests may replace the dip at the cost of harder calibration.

Why this step: v1 said “project onto leading directions and dip,” which is ambiguous for $d_i > 1$ and silently favorable if the direction is tuned on the same data.

3.6 A.6: Selection-aware bootstrap confirmation

When the same sample informs anchor, scale, or projection choices, confirmation must reproduce the complete search. Define

$$T_i^{\text{obs}} = \mathbf{1}\{i \in F^{\text{obs}}\} \text{Dip}\left\{u_i^{\text{obs}\top} \tilde{\psi}_{ij}^{(\star, \text{obs})}\right\}_{j=1}^m,$$

with u_i^{obs} the specified projection of §3.5; setting the statistic to zero for unselected anchors makes screening part of the statistic. For $b = 1, \dots, B$: generate a complete synthetic audit from $\hat{\mathcal{P}}_0$; re-estimate the noise model and whitening; recompute dispersion curves and scale statistics; refit the cross-fitted nulls; rerun screening to get $F^{(b)}$; re-derive projections; recompute $T_i^{(b)}$. Every data-adaptive choice — trimming fractions, fold assignments, K -selection rules, projection pursuit if used — is repeated inside each replicate. The anchor-level bootstrap value is

$$\hat{p}_i^{\text{sel}} = \frac{1 + \sum_{b=1}^B \mathbf{1}\{T_i^{(b)} \geq T_i^{\text{obs}}\}}{B + 1},$$

which has the correct finite-simulation form (and cannot return zero); its calibration is *conditional on the adequacy of the fitted null generator* — this is an approximate parametric bootstrap, not an exact test. The **primary existence claim** uses the omnibus statistics

$$T_{\max}^{\text{obs}} = \max_i T_i^{\text{obs}}, \quad T_{\max}^{(b)} = \max_i T_i^{(b)},$$

whose tail probability automatically absorbs the search over anchors and avoids asserting anchor-level FDR before the null machinery is mature. Resampling only selected projections, or holding σ_i^* , u_i , or F fixed after learning them, does not correct selection. A held-out $\mathcal{S}_2/\mathcal{S}_4$ confirmation remains the cleanest design when the budget allows.

4 Stage B.1: query-level local decomposition

Stage B runs only at confirmed anchors, on the recovery sample $\mathcal{S}_3$, in whitened coordinates. Conditional on its perturbation, query j has one latent label $a_{ij} = g(x_i + \delta_{ij})$, even though the profile is an admixture.

4.1 Local mixtures and soft responsibilities

At $i \in F$, project the recovery sample to d_i leading whitened directions and fit mixtures for $K_i^{\text{loc}} \in \{1, \dots, K_{\max}\}$, selected by held-out likelihood, bootstrap cluster stability, a complexity criterion, and a minimum-mass rule. Retain the posterior responsibilities

$$\gamma_{ij}(\ell) = \hat{\mathbb{P}}(a_{ij} = \ell \mid \tilde{\psi}_{ij}),$$

component centroids $\hat{\mu}_{i\ell}$ and covariances $\hat{C}_{i\ell}$, and estimate compositions *softly*:

$$\hat{\pi}_i(\ell) = \frac{1}{m} \sum_{j=1}^m \gamma_{ij}(\ell).$$

Why this step: hard-assignment frequencies are biased toward $1/K$ under component overlap; responsibilities are the consistent estimator and propagate assignment uncertainty downstream. The minimum-mass rule uses soft mass, $m \hat{\pi}_i(\ell) \geq n_{\min}$. The unperturbed response $\tilde{\psi}_{i0}$ is assigned by posterior to a local component ($\hat{z}_i^{\text{loc}}$), identifying which branch owns the anchor; it is never treated as a pure global archetype.

5 Stage B.2: local-to-global alignment as partial-permutation synchronization

Local labels are arbitrary, and — because not every regime appears in every neighborhood — the alignment problem is synchronization over *partial* matchings, which is harder than synchronization over complete permutations. The committed algorithm:

SYNC1. Graph. Build the anchor graph G from geographic, manifold, or task-metric neighborhoods.

SYNC2. Edge costs. For edge (i, j) and components (ℓ, r) ,

$$c_{ij}(\ell, r) = (\hat{\mu}_{i\ell} - \hat{\mu}_{jr})^\top (\hat{C}_{i\ell} + \hat{C}_{jr} + \lambda I)^{-1} (\hat{\mu}_{i\ell} - \hat{\mu}_{jr}),$$

optionally augmented with local-slope and held-out predictive terms.

SYNC3. Partial matching. Solve a rectangular assignment with an unmatched (dummy) option of cost χ , so components may remain unmatched. Default χ : a high quantile (e.g. 0.95) of same-branch matching costs simulated from the fitted within-regime continuity model, with sensitivity reported.

SYNC4. Edge confidence. Confidence of edge (i, j) is the margin between its best and second-best total assignment cost.

SYNC5. Spanning forest and propagation. Take the maximum-confidence spanning forest of G (Kruskal on edge confidence) and propagate labels along it to define provisional global classes.

SYNC6. Cycle audit and abstention. Every off-forest edge is re-matched independently; the component-level cycle-consistency rate ρ_{cc} is the confidence-weighted fraction of off-forest edges agreeing with the forest labeling. Components with ρ_{cc} below threshold are abstained from, not forcibly synchronized.

SYNC7. Global count. $\widehat{K}$ is the number of global classes whose aggregate soft mass exceeds a floor *and* whose class survives in a stated fraction of Stage-B bootstrap replicates (§6.2).

Why this step: v1 said “synchronize to cycle consistency,” a problem statement rather than a procedure; the forest-plus-audit construction is implementable, and its failure mode (low ρ_{cc}) is an explicit abstention trigger.

Raw global clustering remains rejected: within one regime the response varies across the input space ($w_k(x_i) \neq w_k(x_j)$), so global clusters follow magnitude and covariates rather than mechanism; the local-to-global route follows smooth branches through overlapping neighborhoods. The three-regime Slack-style stack illustrates assembly: local views $\{1, 2\}$, $\{1, 3\}$, $\{2, 3\}$ under different probe families chain into one three-regime system through shared components, even if no unperturbed anchor sits in regime 3.

6 Stage B.3: gates, uncertainty, and certified propagation

6.1 Regularized, responsibility-weighted gate fits

At a flagged anchor with a locally binary boundary, fit

$$\mathbb{P}(a_{ij} = \ell \mid \delta_{ij}) = \text{logit}^{-1}(\beta_{i0} + \beta_i^\top \delta_{ij})$$

by minimizing the cross-entropy against the *soft responsibilities* $\gamma_{ij}(\ell)$ (equivalently, weighted logistic regression), with ridge regularization by default and Firth penalization for small local samples.

Why this step: when clusters separate cleanly in δ -space — the good case — the unpenalized MLE diverges; and hard labels overstate the information available. Then $\hat{\nu}_i = \hat{\beta}_i / \|\hat{\beta}_i\|$ estimates the local boundary normal and the intercept its signed displacement; sparse fits identify candidate gating features; multiclass local fits serve junctions.

6.2 Stage-repeating uncertainty

Ordinary logistic standard errors ignore mixture uncertainty, synchronization error, projection and scale selection, and label-assignment uncertainty. Gate (and composition, and $\hat{K}$) uncertainty is therefore reported from a **Stage-B bootstrap** that resamples the recovery queries and repeats, per replicate: mixture fitting and K_i^{loc} selection, responsibility computation, partial matching, synchronization, and gate fitting. Reported objects: normal cones for $\hat{\nu}_i$, intervals for offsets and $\hat{\pi}_i$, persistence frequencies for global classes.

6.3 Certified propagation: unflagged is not interior

An unflagged anchor may be truly interior — or underpowered, probed in the wrong directions, near a gate with overlapping outputs, part of a rare regime, or a multiplicity miss. Non-rejection is not evidence that no boundary exists, so labels are *not* propagated through unflagged regions by default; doing so would confidently fill exactly the regions where detection lacked sensitivity. Instead, each graph edge used for propagation must be **certified** by bridge probes:

- CERT1.** Place bridge points along the segment (or manifold path) between neighboring anchors x_i, x_j , and probe each with a small battery at a bridge scale.
- CERT2.** The edge is *certified same-branch* if every bridge point passes the minimum-signal and dispersion checks with no modality evidence, and the unperturbed bridge responses interpolate within the posterior predictive tube of a single-branch fit joining the endpoint components.
- CERT3.** The edge is *certified crossing* if an interior bridge point is flagged; bisection along the edge then localizes the boundary, feeding Stage C geometry.
- CERT4.** Otherwise the edge is *uncertifiable*; any label carried across it is reported as EXTRAPOLATED, severed from certified claims.

Propagation abstains in graph components disconnected from any reliably labeled anchor and wherever multiple global labels remain plausible. Every reported regime map distinguishes CERTIFIED from EXTRAPOLATED territory.

Why this step: this replaces v1’s “propagate into unflagged interior regions,” which treated absence of evidence as evidence of absence.

7 Complete annotated algorithm

Stage A: detection.

- A1. Design.** Fix ψ , probe families, ladder, anchors; split the budget $\mathcal{S}_1$ – $\mathcal{S}_4$; register negative controls and support diagnostics. *Every later claim is relative to these choices; registering them first prevents post-hoc tuning.*

- A2. Noise and whitening.** Estimate $\hat{\Sigma}_{\text{obs}}$ from repeats (shrinkage if V large); whiten. *Makes the noise floor isotropic so eigenvalue excess is interpretable.*
- A3. Dispersion curves.** Compute $r_i(\sigma_t)$ on $\mathcal{S}_1$; apply the minimum-signal rule. *Separates “flat/no information” from “null-consistent.”*
- A4. Null generator.** Fit the hierarchical GP $\hat{\mathcal{P}}_0$ robustly (trimmed/calibration-split), with cross-fitting folds for testing. *The null is a fitted population of smooth behaviors, so it must be estimated without contamination from the anchors being tested.*
- A5. Scale statistic and screening.** Compute T_i^{scale} (or R_i^{log}); simulate cross-fitted screening p -values; BH at rate q as a screening device to form F . *Concentrates the confirmation budget; carries no final FDR claim.*
- A6. Shape corroboration.** On $\mathcal{S}_2$, dip test on the specified projections at σ_i^* . *A bump alone is not evidence of populations; modality is the corroborating signature.*
- A7. Confirmation.** Omnibus selection-aware bootstrap for the existence claim; per-anchor $\hat{p}_i^{\text{sel}}$ with dependence-robust correction for anchor-level claims. *All adaptive choices are inside the replicate; calibration is conditional on $\hat{\mathcal{P}}_0$.*

Stage B: recovery (confirmed anchors only).

- B1. Recovery sample.** Fresh queries near σ_i^* from $\mathcal{S}_3$. *Severs recovery from selection.*
- B2. Local mixtures.** Fit, select K_i^{loc} , retain responsibilities; soft compositions $\hat{\pi}_i$. *Soft estimates are consistent under overlap and carry uncertainty forward.*
- B3. Anchor orientation.** Posterior-assign $\tilde{\psi}_{i0}$. *Identifies the branch owning the anchor without archetype assumptions.*
- B4. Synchronization.** SYNC1–7: partial matching with dummy cost, confidence forest, cycle audit, abstention, $\hat{K}$. *Turns arbitrary local labels into global identities with an explicit failure signal.*
- B5. Gates.** Responsibility-weighted ridge/Firth logistic; sparse variant for gating features. *Regularization survives clean separation; weights carry mixture uncertainty.*
- B6. Certification.** Bridge-probe certification of every propagation edge; bisection localizes crossings. *Non-rejection at an anchor never licenses propagation through it.*
- B7. Propagation.** Labels flow only across certified same-branch edges; all else EXTRAPOLATED or abstained. *Certified and extrapolated territory are never mixed in one claim.*
- B8. Stage-B bootstrap.** Repeat B2–B7 per replicate for uncertainty on $\hat{\pi}_i$, $\hat{\nu}_i$, $\hat{K}$, and label persistence. *Uncertainty must include every estimated layer, not only the last one.*

Stage C: interpretation.

- C1. Mechanism characterization.** Features, thresholds, or boundary maps predicting recovered labels and normals, with uncertainty. *Converts geometry into a candidate mechanism.*

- C2. Consequential association.** Association of routing with protected attributes, geography, outcomes; spatially aware inference. *The audit question is who is affected, not only whether structure exists.*
- C3. Probe-support comparison.** Contrast on-manifold, ambient, and feature-targeted findings, stratified by support realism. *Different supports see different edges of the same system.*
- C4. Report.** Existence, structure, interpretation stated separately with abstentions and controls. *Later-stage failure does not erase earlier-stage evidence.*

8 Stage B assumptions and diagnostics

Fix an anchor x_i ; write $Y_{ij} = \psi(f(x_i + \delta_{ij}))$, hidden labels $A_{ij} = g(x_i + \delta_{ij})$, compositions $\pi_i(k)$, and regime-conditioned laws P_{ik} . Recovery is not assumption-free; each assumption is paired with a diagnostic and an abstention condition.

B1 (stationarity). One stable routing law governs the audit: $A_{ij} = g(x_i + \delta_{ij})$, with any randomness stationary. Rules out a stateful adversary recognizing audit traffic. *Diagnostic:* interleave disguised and audit queries; test batchwise drift; withhold static recovery under material drift.

B2 (probe accessibility). Every recovered regime is reached by some probe family: $\exists i : \pi_i(k) \geq \pi_{\min} > 0$. $\widehat{K}$ counts observable, probe-accessible regimes, not internal code paths. *Diagnostic:* compare scales and supports; label support-specific modes as such.

B3 (local distinguishability). Locally present regimes differ in law: $P_{ik} \neq P_{i\ell}$, with bounded optimal local classification error $\inf_h \mathbb{P}(h(Y_{ij}) \neq A_{ij} \mid x_i) \leq \epsilon_{\text{loc}}$. No Gaussian assumption is intrinsic. *Diagnostic:* held-out likelihood, stability, separation statistics; decline decomposition when unstable.

B4 (prevalence). $m \pi_i(k) \geq n_{\min}$ (soft mass). Rare components are not practically recoverable. *Diagnostic:* report soft masses and CIs; merge or omit failing components.

B5 (within-regime continuity). $d_{\mathcal{P}}(P_{ik}, P_{jk}) \leq L d_{\mathcal{X}}(x_i, x_j)$ for neighbors on one branch; permits drifting $w_k(x)$, forbids nothing about global variation. *Diagnostic:* smoothness of matched centroids/covariances along the graph.

B6 (matching margin). $d_{\mathcal{P}}(P_{ik}, P_{jk}) + \gamma < \min_{\ell \neq k} d_{\mathcal{P}}(P_{ik}, P_{j\ell})$, $\gamma > 0$; otherwise branches can cross and exchange identities unobserved. *Diagnostic:* report best-vs-second-best margins, match-stability bootstrap, ρ_{cc} .

B7 (overlap connectivity). Local views must chain (e.g. $\{1, 2\}, \{2, 3\}, \{3, 4\}$); the anchor-regime bipartite graph must be connected over each recovered family. *Diagnostic:* connectivity and articulation points; deletion stability; disconnected families reported separately.

B8 (anchor fidelity). $A_{i0} = g(x_i)$: the unperturbed query is routed by the same mechanism. Used only for orientation. *Diagnostic:* repeated unperturbed queries consistently join one component.

B9 (gate regularity). Near a binary boundary, a local linear approximation $\beta_{i0} + \beta_i^\top \delta = 0$ is adequate at the probed scale; sparsity is an additional assumption used only when claiming a gating feature. *Diagnostic:* held-out label prediction; linear-vs-nonlinear comparison; spatial coherence of normals.

B10 (transversality). $\nu_i^\top \Sigma_{\text{probe}} \nu_i > 0$; probes parallel to the boundary see nothing. *Diagnostic:* compare supports; negative conclusions stay scale- and support-relative.

Identifiability statement. Under B1–B10 the target is recovery of observable regimes up to global label permutation. Semantic names (“biased,” “innocuous”) are produced by Stage C, never by clustering.

9 Stage C: interpretation and audit finding

Multiple regimes do not imply misconduct; benign mixtures of experts, safety mechanisms, and operational controllers are legitimately gated. Stage C characterizes the mechanism (features, thresholds, boundary maps with uncertainty), tests consequential association (protected attributes, geography, energy burden, service quality, error rates — with spatially aware inference), and compares probe supports with deliberately weakened language: ambient-only evidence is *consistent with* an off-manifold scaffold; on-manifold evidence indicates behavior relevant to the modeled data support; a claim of misconduct requires a separate interpretation and impact analysis beyond either. The reporting contract separates **existence** (calibrated Stage-A evidence), **structure** (regimes, compositions, partition, gates, with uncertainty, certification status, and abstentions), and **interpretation** (the argued connection to a consequential or adversarial mechanism). Failure at a later stage does not erase an earlier result.

10 Validation and abstention contract

Recovery is withheld or qualified when any of the following fail: minimum-signal condition; calibrated screening and omnibus confirmation; cluster stability under bootstrap and held-out prediction; robustness of K_i^{loc} to tuning; component separation relative to within-component covariance; matching margins and cycle consistency ρ_{cc} ; cross-support compatibility of shared components; held-out reconstruction of labels and responses; edge certification along every propagation path; negative-control sanity (controls must not be flagged at rates above their nominal levels); support realism thresholds for any finding claimed to be operationally meaningful.

Reported quantities: omnibus and per-anchor selection-adjusted p -values; insufficient-signal, abstention, and certification rates; cluster stability; component separation; ρ_{cc} ; held-out assignment error; soft-composition intervals; normal cones; $\widehat{K}$ persistence; negative-control outcomes; support realism distributions.

11 Relationship to separable profile-mean recovery

The profile-mean approach factors $M = WH + N$ and identifies simplex vertices from pure or nearly pure columns (SPA with simplex-constrained regression). It requires separability, constant mode signatures w_k , and survives on profile means — averaging away exactly the within-profile multimodality that Stage B consumes. The present route replaces separability with B1–B10, allows

$w_k(x)$ to drift (B5), and moves the difficulty from vertex identification to local mixture estimation plus partial-permutation synchronization. Neither dominates universally; when both structures hold, SPA provides a cross-check, and the evaluation plan includes it as a baseline.

12 Evaluation plan

Suites. (1) smooth single-regime nulls with varying slope and curvature; (2) heteroskedastic and anisotropic-noise single-regime nulls; (3) two-regime threshold gates with controlled jump, noise, spatial dependence; (4) three-regime Slack-style stacks with separate on-manifold and ambient probes; (5) overlapping regimes approaching the distinguishability limit; (6) disconnected coverage and deliberately inconsistent matchings; (7) **misspecification**: heavy-tailed noise, unmodeled heteroskedasticity, kinks, wrong Matérn ν , adversarial smoothing near probe scales; (8) **adaptive adversary probes**: response smoothing, noise inflation, probe-pattern detection (scoped; full treatment deferred); (9) baselines: SPA profile-mean recovery, global clustering, residual-outlier auditing, detector-boundary anomaly flagging.

Metrics. Detection: realized FDR and power against jump size and noise; calibration curves of nominal vs. realized error under correct and misspecified generators; stability comparison of R_i (v1 ratio) vs. $R_i^{\log}$ vs. T_i^{scale} ; negative-control flag rates. Recovery: query-label accuracy up to permutation; $\|\hat{\pi}_i - \pi_i\|$; partition accuracy on certified territory; $\widehat{K}$ distribution; boundary-normal angular error; certification precision (rate at which certified same-branch edges truly contain no crossing). Abstention: **operating characteristics** — abstention rate in non-identifiable settings vs. identifiable ones; the method should abstain where it should and rarely where it should not.

13 Bringing Stage B to Stage A’s level

Stage A now has a precise null claim, a concrete calibration route, a stabilized statistic, and a falsification plan. Stage B reaches the same standard when it has:

- P1. A formal estimand and loss.** The target: the restriction of the partition $\{g^{-1}(k)\}$ to probe-accessible territory, up to label permutation; the composition field π_i ; local normals ν_i . Losses: permutation-minimized partition error on certified territory, $\|\hat{\pi}_i - \pi_i\|$, angular normal error. Stage A has a hypothesis; Stage B needs an estimand of the same precision.
- P2. Fully committed procedures.** This note commits the synchronization (SYNC1–7) and certification (CERT1–4) algorithms; remaining freedom — χ , confidence thresholds, $n_{\min}$, bridge densities — needs registered defaults with sensitivity analyses, so that two auditors given the same data produce the same output.
- P3. Uncertainty that propagates every layer.** The Stage-B bootstrap of §6.2 must cover mixture fitting, model selection, matching, synchronization, and gate fitting jointly; reported intervals that skip a layer are understatements.
- P4. A theorem-shaped target.** Minimal version: under B3–B7 with margin γ , matching-cost concentration, and per-edge certified-matching error α_e , forest propagation recovers global labels up to permutation with probability at least $1 - \sum_{e \in \text{forest}} \alpha_e$; local clustering error enters α_e through existing finite-sample mixture bounds; soft compositions satisfy a CLT under the local model. None of this is claimed proven here; it is the target that makes Stage B a result rather than a pipeline.

- P5. Abstention operating characteristics.** Quantitative evidence, from suite (6) and the misspecification suite, that the diagnostics abstain when assumptions fail — “knows when it does not know” as a measured property, not a design intention.
- P6. Inherited selection discipline.** Stage B runs on selected anchors; estimates reusing selection data inherit winner’s-curse bias. The budget split $(\mathcal{S}_3, \mathcal{S}_4)$ is the design-level fix; any reuse must enter the Stage-B bootstrap.

14 Claim boundary

This note specifies a complete, revised algorithmic proposal. The central research risk is no longer whether the recovery idea makes sense; it is whether Stage A can make a precise, calibrated claim that distinguishes boundary-like latent behavior from ordinary smooth complexity, heteroskedasticity, and null-model misspecification. That is addressed by the concrete null generator, cross-fitted and simulation-calibrated inference, the stabilized scale statistic, and the falsification suites — and it must be demonstrated empirically before Stage A is presented as a result. Stage B is presented honestly as a partially specified structural-recovery layer whose outputs are reported only where synchronization, certification, and gate diagnostics pass, with §13 as its maturation contract. Detection remains the most developed component; recovery is the proposed next stage; interpretation is a substantive argument, not an algorithmic output.